

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Ethical Hacking

COURSE CODE: ITA1407

COURSE OUTCOME COVERED

CO4: Implement network security testing methods by performing web server attacks, analyzing web application vulnerabilities, and assessing wireless network security using Kali Linux.

1. TJX Companies Wireless Network Breach, 2005–2007

TJX Companies experienced a major payment-card data breach after attackers obtained unauthorized wireless access to its retail network. Investigations found that TJX had used WEP while transitioning to WPA and had not implemented adequate controls to restrict wireless access. The incident demonstrated the risks of obsolete encryption, inadequate monitoring, weak network segmentation, and insecure transmission of sensitive data.

Assignment Question: Prepare a Vulnerability Assessment Report examining the wireless encryption, authentication, network segmentation, monitoring, and payment-data protection weaknesses that contributed to the TJX incident. Recommend how WPA2/WPA3, enterprise authentication, rogue-access-point detection, and PCI DSS controls could have reduced the risk.

COMMANDS AND OUTPUT:

1. ip link

```
(kali㉿kali)-[~]  
$ ip link  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000  
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000  
   link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff
```

2. nmcli device status

```
(kali㉿kali)-[~]  
$ nmcli device status  
DEVICE  TYPE      STATE              CONNECTION  
eth0    ethernet  connected          Wired connection 1  
lo      loopback  connected (externally)  lo
```

3. ip addr

```
(kali㉿kali)-[~]  
$ ip addr  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff  
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0  
        valid_lft 85367sec preferred_lft 85367sec  
    inet6 fd17:625c:f037:2:50f0:f698:66ab:20d5/64 scope global dynamic noprefixroute  
        valid_lft 86135sec preferred_lft 14135sec  
    inet6 fe80::e98b:3c40:9f97:8af0/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever
```

4. ip route

```
(kali㉿kali)-[~]  
$ ip route  
default via 10.0.2.2 dev eth0 proto dhcp src 10.0.2.15 metric 100  
10.0.2.0/24 dev eth0 proto kernel scope link src 10.0.2.15 metric 100
```

5. ip neigh

```
(kali㉿kali)-[~]  
$ ip neigh  
10.0.2.2 dev eth0 lladdr 52:54:00:12:35:00 STALE  
fd17:625c:f037:2::2 dev eth0 lladdr 52:54:00:12:35:00 router STALE  
fe80::2 dev eth0 lladdr 52:54:00:12:35:00 router STALE  
fd17:625c:f037:2::3 dev eth0 lladdr 52:54:00:12:35:00 router STALE
```

6. ss -tuln

```
(kali㉿kali)-[~]  
$ ss -tuln  
Netid      State      Recv-Q     Send-Q     Local Address:Port     Peer Address:Port
```

7. nmap -sn 10.0.2.0/24

```
(kali@kali)-[~]
$ nmap -sn 10.0.2.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-20 01:25 -0400
Stats: 0:00:01 elapsed; 0 hosts completed (0 up), 255 undergoing ARP Ping Scan
ARP Ping Scan Timing: About 52.74% done; ETC: 01:25 (0:00:02 remaining)
Stats: 0:00:01 elapsed; 0 hosts completed (0 up), 255 undergoing ARP Ping Scan
ARP Ping Scan Timing: About 55.10% done; ETC: 01:25 (0:00:02 remaining)
Stats: 0:00:01 elapsed; 0 hosts completed (0 up), 255 undergoing ARP Ping Scan
ARP Ping Scan Timing: About 62.74% done; ETC: 01:25 (0:00:01 remaining)
Packet Tracing enabled.
SENT (1.6789s) ARP who-has 10.0.2.2 tell 10.0.2.15
SENT (1.6806s) ARP who-has 10.0.2.81 tell 10.0.2.15
SENT (1.6815s) ARP who-has 10.0.2.82 tell 10.0.2.15
SENT (1.6823s) ARP who-has 10.0.2.83 tell 10.0.2.15
SENT (1.6825s) ARP who-has 10.0.2.84 tell 10.0.2.15
SENT (1.6826s) ARP who-has 10.0.2.85 tell 10.0.2.15
SENT (1.6827s) ARP who-has 10.0.2.86 tell 10.0.2.15
SENT (1.6828s) ARP who-has 10.0.2.87 tell 10.0.2.15
SENT (1.6829s) ARP who-has 10.0.2.88 tell 10.0.2.15
SENT (1.6830s) ARP who-has 10.0.2.89 tell 10.0.2.15
SENT (1.6831s) ARP who-has 10.0.2.90 tell 10.0.2.15
SENT (1.6832s) ARP who-has 10.0.2.91 tell 10.0.2.15
SENT (1.6833s) ARP who-has 10.0.2.94 tell 10.0.2.15
SENT (1.6834s) ARP who-has 10.0.2.95 tell 10.0.2.15
SENT (1.6835s) ARP who-has 10.0.2.96 tell 10.0.2.15
SENT (1.6838s) ARP who-has 10.0.2.97 tell 10.0.2.15
SENT (1.6839s) ARP who-has 10.0.2.98 tell 10.0.2.15
SENT (1.6840s) ARP who-has 10.0.2.101 tell 10.0.2.15
SENT (1.6841s) ARP who-has 10.0.2.102 tell 10.0.2.15
SENT (1.6842s) ARP who-has 10.0.2.103 tell 10.0.2.15
SENT (1.6843s) ARP who-has 10.0.2.104 tell 10.0.2.15
SENT (1.6844s) ARP who-has 10.0.2.105 tell 10.0.2.15
SENT (1.6845s) ARP who-has 10.0.2.106 tell 10.0.2.15
SENT (1.6846s) ARP who-has 10.0.2.107 tell 10.0.2.15
SENT (1.6847s) ARP who-has 10.0.2.110 tell 10.0.2.15
SENT (1.6848s) ARP who-has 10.0.2.111 tell 10.0.2.15
SENT (1.6849s) ARP who-has 10.0.2.112 tell 10.0.2.15
SENT (1.6850s) ARP who-has 10.0.2.113 tell 10.0.2.15
SENT (1.6851s) ARP who-has 10.0.2.114 tell 10.0.2.15
SENT (1.6852s) ARP who-has 10.0.2.115 tell 10.0.2.15
SENT (1.6853s) ARP who-has 10.0.2.116 tell 10.0.2.15
SENT (1.6854s) ARP who-has 10.0.2.117 tell 10.0.2.15
SENT (1.6854s) ARP who-has 10.0.2.118 tell 10.0.2.15
SENT (1.6859s) ARP who-has 10.0.2.119 tell 10.0.2.15
SENT (1.6859s) ARP who-has 10.0.2.120 tell 10.0.2.15
SENT (1.6860s) ARP who-has 10.0.2.121 tell 10.0.2.15
SENT (1.6861s) ARP who-has 10.0.2.122 tell 10.0.2.15
SENT (1.6862s) ARP who-has 10.0.2.123 tell 10.0.2.15
SENT (1.6863s) ARP who-has 10.0.2.126 tell 10.0.2.15
```

8. nmap -sV 10.0.2.15

```
(kali@kali)-[~]
$ nmap -sV 10.0.2.15
Nmap 7.99 ( https://nmap.org )
Usage: nmap [Scan Type(s)] [Options] {target specification}
TARGET SPECIFICATION:
  Can pass hostnames, IP addresses, networks, etc.
  Ex: scanme.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255.1-254
  -iI <inputfilename>: Input from list of hosts/networks
  -iR <num hosts>: Choose random targets
  --exclude <host1[,host2][,host3],...>: Exclude hosts/networks
  --excludefile <exclude_file>: Exclude list from file
HOST DISCOVERY:
  -sL: List Scan - simply list targets to scan
  -sn: Ping Scan - disable port scan
  -Pn: Treat all hosts as online -- skip host discovery
  -PS/PA/PY/PP[<portlist>]: TCP SYN, TCP ACK, UDP or SCTP discovery to given ports
  -PE/PP/PM: ICMP echo, timestamp, and netmask request discovery probes
  -PO[<protocol list>]: IP Protocol Ping
  -n/-R: Never do DNS resolution/Always resolve [default: sometimes]
  --dns-servers <serv1[,serv2],...>: Specify custom DNS servers
  --system-dns: Use OS's DNS resolver
  --traceroute: Trace hop path to each host
SCAN TECHNIQUES:
  -sS/sT/sA/sW/sM: TCP SYN/Connect()/ACK/Window/Maimon scans
  -sU: UDP Scan
  -sN/sF/sX: TCP Null, FIN, and Xmas scans
  --scanflags <flags>: Customize TCP scan flags
  -sI <zombie host[:probeport]>: Idle scan
  -sV/sZ: SCTP INIT/COOKIE-ECHO scans
  -sO: IP protocol scan
  -b <FTP relay host>: FTP bounce scan
PORT SPECIFICATION AND SCAN ORDER:
  -p <port ranges>: Only scan specified ports
  Ex: -p22; -p1-65535; -p U:53,111,137,T:21-25,80,139,8080,S:9
  --exclude-ports <port ranges>: Exclude the specified ports from scanning
  -F: Fast mode - Scan fewer ports than the default scan
  -r: Scan ports sequentially - don't randomize
  --top-ports <number>: Scan <number> most common ports
  --port-ratio <ratio>: Scan ports more common than <ratio>
SERVICE/VERSION DETECTION:
  -sV: Probe open ports to determine service/version info
  --version-intensity <level>: Set from 0 (light) to 9 (try all probes)
  --version-light: Limit to most likely probes (intensity 2)
  --version-all: Try every single probe (intensity 9)
  --version-trace: Show detailed version scan activity (for debugging)
SCRIPT SCAN:
  -sC: equivalent to --script-default
  --script=<Lua scripts>: <Lua scripts> is a comma separated list of
    directories, script-files or script-categories
  --script-args=<n1=v1[,n2=v2,...]>: provide arguments to scripts
```

9. ip -br link

```
(kali㉿kali)-[~]  
$ ip -br link  
lo          UNKNOWN      00:00:00:00:00:00 <LOOPBACK,UP,LOWER_UP>  
eth0       UP            08:00:27:5a:87:bc <BROADCAST,MULTICAST,UP,LOWER_UP>
```

10. mkdir -p ~/Q1_evidence ip link > ~/Q1_evidence/interfaces.txt nmcli device status > ~/Q1_evidence/device-status.txt ip addr > ~/Q1_evidence/ip-address.txt ip route > ~/Q1_evidence/routes.txt ip neigh > ~/Q1_evidence/neighbors.txt ss -tuln > ~/Q1_evidence/connections.txt nmap -sn 10.0.2.0/24 > ~/Q1_evidence/network-scan.txt

```
(kali㉿kali)-[~]  
$ mkdir -p ~/Q1_evidence 66 ip link > ~/Q1_evidence/interfaces.txt 66 nmcli device status > ~/Q1_evidence/device-status.txt 66 ip addr > ~/Q1_evidence/  
ip_address.txt 66 ip route > ~/Q1_evidence/routes.txt 66 ip neigh > ~/Q1_evidence/neighbors.txt 66 ss -tuln > ~/Q1_evidence/connections.txt 66 nmap -sn 1  
0.0.2.0/24 > ~/Q1_evidence/network-scan.txt
```

11. ls -l ~/Q1_evidence

```
(kali㉿kali)-[~]  
$ ls -l ~/Q1_evidence  
total 28  
-rw-rw-r-- 1 kali kali  63 Aug 20 01:33 connections.txt  
-rw-rw-r-- 1 kali kali 186 Aug 20 01:33 device-status.txt  
-rw-rw-r-- 1 kali kali 331 Aug 20 01:33 interfaces.txt  
-rw-rw-r-- 1 kali kali 851 Aug 20 01:33 ip_address.txt  
-rw-rw-r-- 1 kali kali 242 Aug 20 01:33 neighbors.txt  
-rw-rw-r-- 1 kali kali 396 Aug 20 01:33 network-scan.txt  
-rw-rw-r-- 1 kali kali 138 Aug 20 01:33 routes.txt
```